

Expression of Interest (EOI) and Ideation Proposal

Smart Physiotherapy Monitoring Sleeve

Team Name: PhysioPulse

A. Statement of Interest

We are pleased to submit this Expression of Interest for the IEEE MYOSA contest, proposing the development of a Smart Physiotherapy Monitoring Sleeve. Physiotherapy is a critical part of recovery from surgery, injury, and neurological conditions, yet a persistent gap exists between what a physiotherapist prescribes and what a patient actually performs once they leave the clinic. Repetitions, form, tempo, pain, and recovery trends are presently captured only through subjective, memory-based verbal reporting, which often leads to inaccurate or delayed clinical insight. Our team believes that the sensor suite available on the MYOSA kit is exceptionally well suited to closing this gap through a single, repositionable wearable sleeve capable of objectively tracking repetitions, range of motion, exercise tempo, pain events, falls in real time. We are motivated by a real clinical scenario in which a physiotherapist has no reliable way to verify home exercise compliance, and we view this contest as an opportunity to design, build, and demonstrate a working prototype that brings genuine, measurable value to patients – particularly the elderly and those recovering from stroke, surgery, or Parkinson's disease. We look forward to presenting a complete, functioning demonstration at the selected conference venue.

B. Details of Team Members

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Ideation Details

How the MYOSA Board Will Be Used in the Project

The MYOSA kit, built around an ESP32 microcontroller, will serve as the central processing and communication unit of the wearable sleeve. It will:

- Connect to the physiotherapist's device over WiFi to receive session prescriptions (exercise type, target repetitions, and patient ID).
- Continuously sample the onboard inertial and environmental sensors over the I2C bus during the session.
- Run all on-board detection logic – repetition counting, tempo classification, fall detection in real time.
- Transmit pain-event alerts, fall alerts, and the final structured session report back to the doctor's device over WiFi.

Sensors and Actuators to Be Used

- **MPU6050 (Accelerometer + Gyroscope):** Core motion sensor used for repetition counting, joint-angle and range-of-motion tracking, exercise tempo (angular velocity), fall detection, step counting and gait cadence.
- **APDS9960 (Gesture Sensor):** Enables button-free, real-time pain reporting; the patient swipes a free hand in front of the sensor to log a pain event with exercise name, repetition number, and timestamp.
- **BMP180 (Pressure Sensor):** Its onboard temperature reading is monitored continuously against the patient's resting baseline as a body-temperature warning check; a sustained or rapid rise beyond the baseline is logged as a "Possible Fever" warning sign in the session report for the doctor to review (used as a relative indicator, not a calibrated clinical thermometer).
- **OLED Display :** Used as an on-sleeve actuator that gives the patient direct, real-time feedback – displaying the live valid repetition count (e.g., "Rep 7/15"), simple session status messages, and pain/fall acknowledgements, so the patient does not need to rely on the doctor's separate device to know their progress during the session

Additional Circuit and/or Software to Be Built Around the MYOSA Board

- **Firmware (ESP32/Arduino C++):** WiFi prescription-receive module, a 100 Hz sensor-polling loop, a threshold based repetition and fall detection state machine.
- **Battery and Battery Module:** A compact 3.7 V Li-Po cell (500–1000 mAh) paired with a TP4056-based charging/protection module regulates power to the ESP32 motherboard and all sensors, allowing the sleeve to run fully untethered during arm or walking exercises. This gives roughly 3–5 hours of continuous monitoring per charge sufficient for a full physiotherapy session and the cell is recharged via USB between sessions.
- **Wearable enclosure:** A lightweight, repositionable fabric sleeve housing the MYOSA board and sensors, wearable on the forearm/upper arm for arm exercises or the ankle/lower leg for walking exercises.
- **Doctor-facing dashboard (software application):** A lightweight web or mobile interface allowing the physiotherapist to send prescriptions, receive real time fall/pain alerts, and view structured end-of-session reports.
- **Data storage and trend-analysis layer:** A simple backend/database to store session reports across days and compute the day-by-day condition assessment (Improving / Stable / Needs Attention / Excellent Progress) using rule-based comparison of valid-rep score, pain-event count.
- **Buzzer:** Sounds an immediate audible alert on the sleeve itself the moment a fall is detected.

Significance of the Application

Millions of patients undergo physiotherapy weekly, but recovery is currently tracked only through subjective, memory-based reporting once the patient leaves the clinic. Physiotherapists have no objec-

tive way to verify whether prescribed exercises were performed correctly, whether pain occurred, or whether a fall took place at home. This project replaces guesswork with continuous, sensor-driven evidence – automatically counting valid repetitions, logging pain in real time and detecting falls.

Intended Use of the MYOSA Platform and How the Project Exploits Its Potential

This project uses every sensor in the MYOSA kit. The MPU6050 does more than track motion – it counts valid reps, checks exercise speed, and detects falls. The APDS9960 gesture sensor lets patients report pain with a hand swipe, no buttons needed. The BMP180, normally used for room readings, is repurposed to check body temperature for fever warnings. The OLED screen shows patients their live rep count. One simple board delivers a full, low-cost health-monitoring system.

Demonstration Planned at the Conference Venue

At the conference venue, the team will demonstrate a complete live session end-to-end:

- A team member will wear the sleeve on the forearm; a prescription (e.g., 10 arm raises) will be sent wirelessly from a laptop/phone acting as the doctor's device.
- The wearer will perform repetitions of varying quality (full range and partial range) to show live valid/invalid rep classification and tempo flagging on the doctor's screen.
- A pain gesture will be demonstrated to show real time pain event logging with repetition number and timestamp.
- A controlled simulated fall will be performed to trigger and display the live fall alert with the buzzer alarm.
- At session end, the audience will be shown the automatically generated structured session report and a simulated multi day progress view illustrating the automated condition assessment (e.g., Improving / Stable / Needs Attention).